



Project Report

On

DIGITAL VOTING MACHINE

Course No: ECE2104

Course Name: Digital Electronics and Logic Circuits Laboratory

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Abstract

The Digital Voting Machine is a logic-based electronic system designed to record and count votes accurately using digital electronics components. The project is implemented without any microcontroller and is based entirely on combinational and sequential logic circuits. Each candidate is represented by a push button, and when a voter presses a button, the corresponding vote count increases and is displayed through a 7-segment display.

The system also includes a locking mechanism using flip-flops to prevent multiple voting during a single voting cycle. Logic gates are used to control inputs and ensure proper system operation. Counters and decoder ICs are used to process and display the vote count in numerical form.

The project demonstrates the practical application of digital electronics in real-world systems such as electronic voting. It improves understanding of logic gates, counters, flip-flops, timers, and display interfacing. The system provides faster and more reliable vote counting compared to manual methods and reduces the possibility of human error.

Introduction

Digital electronics play a vital role in modern technology and communication systems. From computers and mobile phones to industrial automation and security systems, digital logic circuits are widely used in almost every electronic device. Logic gates, counters, flip-flops, and timers form the foundation of digital systems.

Voting is an important process used in elections, institutions, and organizations to make decisions. Traditional paper-based voting systems are often slow, time-consuming, and prone to counting errors. To overcome these limitations, electronic voting systems are becoming increasingly popular due to their speed, reliability, and efficiency.

The aim of this project is to design and implement a Digital Voting Machine using basic digital electronics components without using any programmable microcontroller. The project applies both combinational and sequential logic circuits to create a secure and efficient voting system. The machine records votes digitally, displays results instantly, and prevents multiple voting through a locking mechanism.

This project also helps develop practical knowledge of digital circuit design, simulation, troubleshooting, and hardware implementation.

Methodology

The Digital Voting Machine was designed and implemented using digital logic circuits and standard TTL ICs. The project consists of several stages including input processing, vote validation, counting, display, and reset operations.

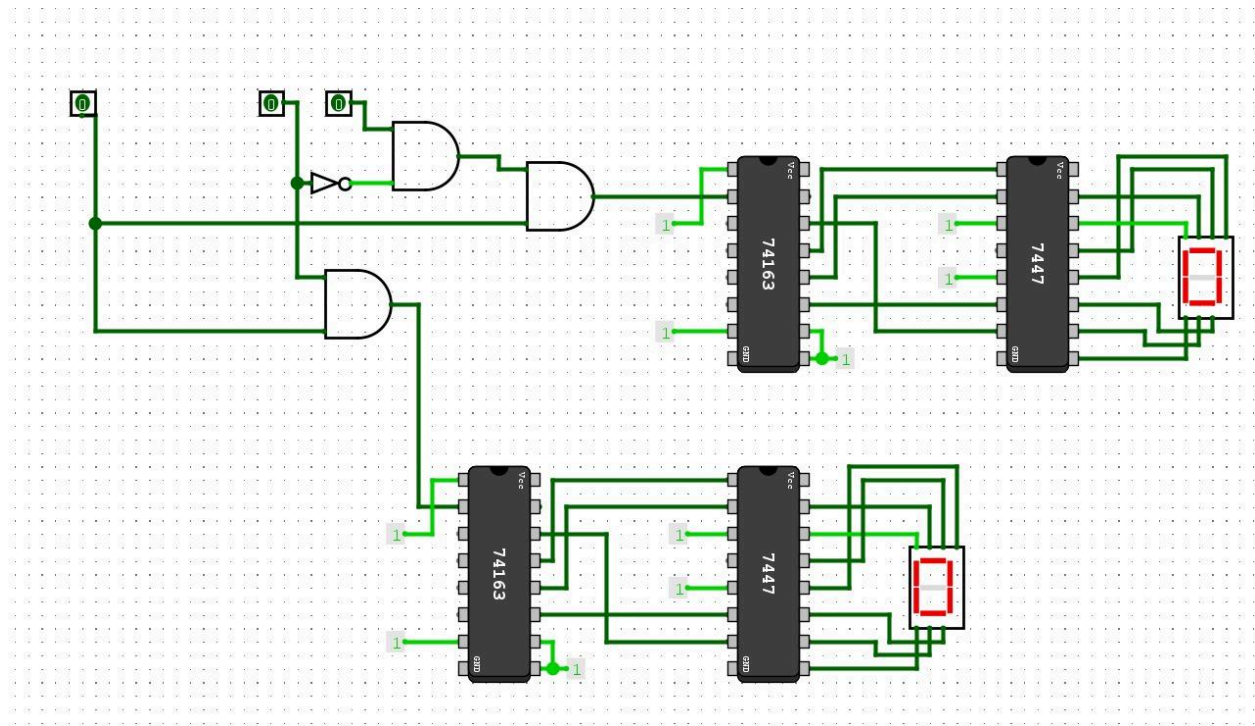
Initially, push buttons are used to represent different candidates. When a voter presses a candidate button, the signal passes through logic gates that verify the validity of the input. AND and OR gates are used to control signal flow and prevent multiple inputs at the same time.

A flip-flop-based locking mechanism is implemented to ensure that only one vote is accepted during a voting cycle. Once a vote is cast, the flip-flop changes state and temporarily disables all other input buttons until the system is reset.

The valid voting signal is then sent to a counter IC such as 7490 or 4026, which increments the vote count. The counter output is connected to a 7447 decoder IC that drives the 7-segment display to show the vote count in decimal form.

The complete circuit was first simulated in Proteus software to verify proper operation before hardware implementation on a breadboard.

Circuit Schematic



Working Principle

1. A voter presses a candidate button.
2. The input passes through logic gates for validation.
3. The flip-flop lock activates after receiving the first valid vote.
4. The counter increments by one.
5. The updated vote count appears on the 7-segment display.
6. The reset button clears all counts and unlocks the system.

Result Analysis

The implemented Digital Voting Machine successfully counted and displayed votes for different candidates. The counters responded accurately to valid button presses, and the 7-segment displays correctly showed the updated vote count.

The flip-flop locking mechanism effectively prevented multiple voting during a single voting cycle. The system also demonstrated stable operation during simulation and hardware testing. Overall, the project achieved the intended objectives and operated according to the proposed specifications.

Problem and Technical Issues Encountered

During implementation, several technical issues were encountered:

- Multiple counts occurred due to switch bouncing during button presses
- Incorrect wiring initially caused display errors
- Noise in the circuit occasionally generated false triggering
- Proper synchronization between counters and displays required careful debugging
- Managing multiple IC connections on the breadboard was challenging

These problems were solved by careful wiring, proper grounding, signal stabilization, and repeated testing.

Conclusion

The Digital Voting Machine was successfully designed and implemented using digital logic circuits without any microcontroller. The project demonstrated the practical use of logic gates, counters, flip-flops, and display systems in creating a secure and efficient electronic voting system.

The system accurately counted votes, displayed results in real time, and prevented multiple voting using a lock mechanism. The project enhanced understanding of digital electronics concepts and provided valuable experience in circuit simulation and hardware implementation.

Although the project achieved its objectives, additional features such as password protection, wireless communication, or larger vote storage could further improve the system in the future.

Future Improvements

- Add password-protected admin access
- Increase number of candidates
- Add buzzer and invalid vote indication
- Implement backup memory system
- Design PCB for compact implementation

GitHub Repository

<https://github.com/tiabaalam/Digital-Voting-Machine-Circuit>